

R.M.S. TITANIC · 1912

WHO SURVIVED?

An Exploratory Data Analysis of 889 passengers

my first internship project!

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INTERN

CODEALPHA

Data Analytics
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ABOUT THIS PROJECT

THE QUESTION What really decided who lived and who did not on the Titanic: gender, class, age, money, or family?

891

passengers

12

variables

38%

survived

TOOLKIT Python · Pandas · NumPy · Matplotlib · Seaborn · Google Colab

MEET THE DATASET

SURVIVED

0 = died, 1 = lived

PCLASS

Ticket class 1, 2, 3

SEX

Male or female

AGE

Years (177 missing)

SIBSP

Siblings / spouse
aboard

PARCH

Parents / children
aboard

FARE

Ticket price (£)

EMBARKED

Port: S, C or Q

CABIN

Room (77% missing)

NAME, TICKET

Identifiers, not
analysed

12 columns in total. Survived is the outcome we want to explain.

CHAPTER III

MY METHOD

1

LOAD

Read the CSV into a
Pandas DataFrame

2

INSPECT

`head()`, `info()`,
`describe()`, `isnull()`

3

CLEAN

Fix missing values,
check duplicates

4

VISUALISE

Count plots,
histogram, heatmap

5

CONCLUDE

Write an insight
under every chart

every step is in my Colab notebook

CLEANING THE DATA

AGE

177 missing

Filled with the median (28 years)

CABIN

687 missing · 77%

Too sparse, so the column was dropped

EMBARKED

2 missing

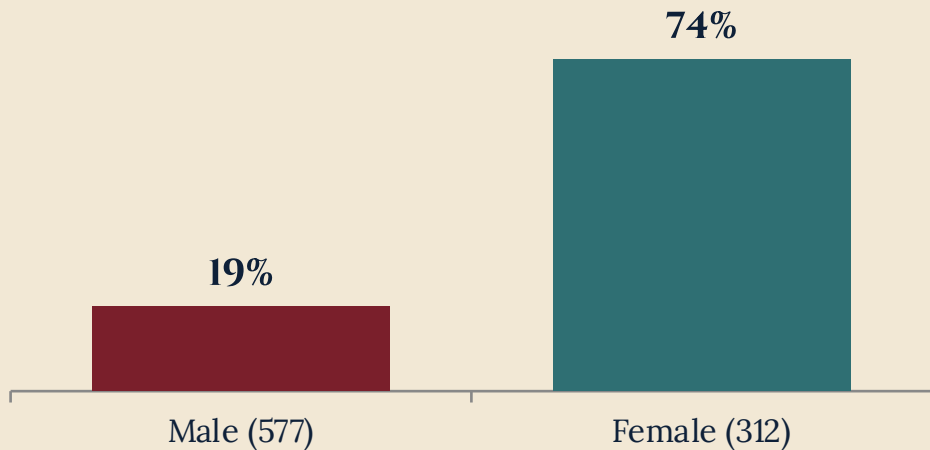
Only 2 rows, so they were removed

AFTER CLEANING: 889 rows · 0 missing · 0 duplicates

FINDING ONE

WOMEN FIRST

SURVIVAL RATE BY SEX



"women and children first" held true

109 OF 577

men survived

231 OF 312

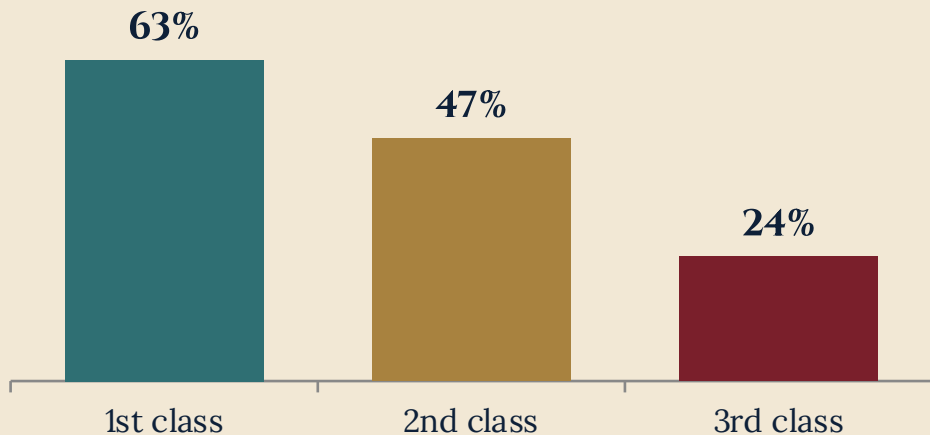
women survived

**Women were nearly 4 times
more likely to live.**

FINDING TWO

MONEY BOUGHT SAFETY

SURVIVAL RATE BY CLASS



MEDIAN FARE PAID

£58.69 1st class

£14.25 2nd class

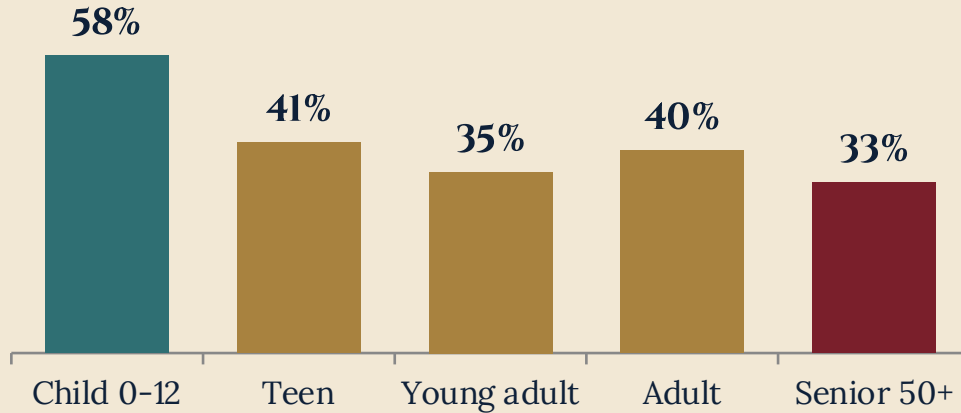
£8.05 3rd class

Pclass vs Survived: $r = -0.34$

FINDING THREE

AGE AND FAMILY

SURVIVAL RATE BY AGE GROUP



FAMILY SIZE

Alone: 30% survived

1 to 3 relatives: 55% to 72%

4 or more: only 16%

Small families helped. Very large ones did not.

CONCLUSION

KEY TAKEAWAYS

- 1 **GENDER** was the strongest factor: 74% of women vs 19% of men survived.
- 2 **CLASS** mattered: 63% in 1st class vs 24% in 3rd.
- 3 **FARE** was linked to survival ($r = +0.26$), mostly because it reflects class.
- 4 **AGE** had a weak overall effect ($r = -0.07$), but children did better.
- 5 **FAMILY** of 1 to 3 helped; very large families fared worst.

EPILOGUE

WHAT I LEARNED

- Handling missing data the right way
- Asking questions before charting
- Reading correlations without over-claiming
- Turning charts into plain-English insights

FULL CODE & NOTEBOOK

GITHUB:

CODEALPHA_EDA

Thank you, @CodeAlpha, for this opportunity!

Next: Visualization & Sentiment Analysis